Teaching Statement

Waseem Raza, PhD | Assistant Professor in Data Science | Seton Hall University

Teaching computing as evidence-based problem solving

My central teaching goal is for students to understand not only how to use a programming language or fit a machine-learning model, but how to explain and test their choices. In a foundational programming course, that means decomposing a problem, tracing code, choosing suitable data structures, and writing tests. In data science, it also means examining data provenance, leakage, sampling bias, model generalization, uncertainty, and the real-world consequences of mistakes. I want students to develop the confidence to ask good questions and the discipline to check their answers.

I have taught undergraduate Computer Programming, Communication Systems, and Wireless Communication at the University of Lahore and supported engineering laboratories and course assessment at the University of Engineering and Technology, Taxila. I have also mentored students in research-oriented simulations, ML implementations, and technical writing through COMSATS and my doctoral research group at the University of Oklahoma. These settings required adapting explanations to students with different mathematical and programming backgrounds. They also reinforced the value of concrete examples followed by independent practice.

Courses and instruction across levels

At Seton Hall, I would be prepared to contribute to introductory programming and computational problem solving, Python programming for data science, data analytics, introductory and applied machine learning, data mining, AI, and project-based courses. My background also supports computer networking and data engineering. For broader undergraduate CS courses such as data structures and algorithms, I would develop and review syllabus-specific material to align with departmental outcomes. I would approach graduate data-science courses through deeper statistical evaluation, reliable learning under dataset shift, scalable analytical workflows, and research-informed projects.

In a programming course, students could begin with short trace-and-predict exercises before implementing functions, tests, and readable programs. In data science, a scaffolded sequence could move from cleaning and visualization to feature engineering, establishing a baseline, selecting metrics, and comparing models on a defensible held-out dataset. In machine learning, students could investigate how class imbalance, changing data distributions, or an unrepresentative test split alter conclusions. I would assess both analytical reasoning and executable implementation, not simply the highest predictive score.

An inclusive and accountable learning environment

Students enter CS and data science through varied routes: some have built programs already, while others may be learning to use a terminal and debug an error for the first time. I would make prerequisites explicit, provide reproducible starter environments, model debugging strategies, and use regular low-stakes checks so misconceptions surface early. Assignments would have clear rubrics that reward reasoning, correct implementation, transparent experiment design, documentation, and respectful teamwork. Team projects would assign rotating technical and explanatory roles, helping prevent prior programming experience from determining who participates.

Teaching data science also entails discussing where datasets come from and whose experiences they represent. When analyzing human activity or service data, students should ask whether a model performs consistently across groups and contexts, whether the data should be used, and how to report meaningful uncertainty. I would integrate these questions into ordinary technical exercises instead of treating ethics as an unrelated topic.

Mentoring through authentic research

Research mentorship should be accessible to students at different stages. A new student might begin by reproducing a simple classifier or producing a well-documented dataset audit. A more advanced student could test a controlled hypothesis about distribution shift, examine false positives in time-series classification, or write a conference-style report. I would begin with a specific question, milestones, baseline, evaluation criteria, and a reproducible repository. Regular meetings and code reviews would support technical progress and scientific judgment, including the ability to report negative results.

My own research and industry experience in wireless network analytics and wearable sensing provide examples, but student projects would use public or synthetic datasets unless institutional permission and data protections were in place. I would also support students preparing independent studies, capstones, conference posters, and graduate research proposals, in coordination with department colleagues.

Continuous improvement and contribution

I would use student work, course-level learning outcomes, and midsemester feedback to revise explanations and assessments. I am interested in contributing to departmental advising, student recruitment, outreach, and cross-disciplinary offerings in Data Science and Engineering. Above all, I want students to leave a course with practical skills, sound mathematical and computational reasoning, and the habits of honesty, curiosity, and collaboration needed to keep learning.